Compositional system

geometry and dynamics for,

Markham municipal operating budget

2011-2025

Report title	Compositional system geometry and dynamics for, Markham municipal operating budge
Generated	2026-05-06
Run ID	v1.1.x-final
Dataset ID	markham_budget
T (records)	15
D (carriers)	8
Engine version	cnt 2.0.4
Schema version	2.1.0
Engine signature (sha256[:32])	97748868a0381cb7b995d2ae562e5c42…
Source CSV SHA-256	dff512242fcb4dac6a99efa41252a8dc…
Closed data SHA-256	8983a79cbf6e0b7b28469884bd9d3ff3…
Content SHA-256 (canonical JSON)	f223ce2f2085f17b90a496c2754570c4…

```
Reading order: geometry first, dynamics second.
  §A  Data disclosure & processing notes
  §B  Geometry : structure on the simplex
              evolution of proportions → ternaries → variation
              → correlation → scree → biplot → top-N biplot
              → balance dendrogram → SBP table
  §C  Dynamics : evolution through compositional space
              trajectories (Hs, ‖clr‖, ILR axes) → System Course Plot
              → bearing rose → bearing heatmap → ω → metric trace
              → helmsman → ring class → locks → CBS
              → pairwise divergence ranking → ω↔Hs
  §D  Summary
```

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Plate 2 | §A Data Disclosure & Processing-Anomaly Review

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

```
INPUT DATA
  source file        : markham_budget_input.csv
  source SHA-256     : dff512242fcb4dac6a99efa41252a8dc407970d51dfecd2299027dc0b802cf04
  closed data SHA-256: 8983a79cbf6e0b7b28469884bd9d3ff3998b0982ec01fa418ec95c1c7bb5936e
  records (T)        : 15
  carriers (D)       : 8
  carriers           : Operations & Asset Mgmt, Public Safety, Planning & Building, Parks, Culture, Library & Heritage, Engineering & Capital, Corporate

ORDERING
  is_temporal        : True
  ordering_method    : by-time

ZERO REPLACEMENT
  zeros replaced     : 0
  replacement δ      : 1e-15
  method             : multiplicative

UNITS LINEAGE  (schema 2.1.0)
  input_units                 : ratio
  higgins_scale_units         : neper
  units_scale_factor_to_neper : 1.0

ENGINE CONFIG ACTIVE OVERRIDES
  (none — defaults from cnt.py USER CONFIGURATION block)
```

```
ENGINE SOURCE FINGERPRINT
  stage2_locked.py      : c318ab82c15890b947c3fc15a6c66b55aa23f3cf94104cf9b9d5fbdb
  cnt.py                : 1e10cb62d6588ac9557c6a2e8436c2980f0c35b379f29c05c0377711
  combined_sha256       : 97748868a0381cb7b995d2ae562e5c429988ee7fd556943b882c5fcb

DEGENERACY FLAGS
  ⚠ {'flag': 'small_T', 'severity': 'warning', 'message': 'Engine: T=15 is small, shortrate
  ⚠ {'flag': 'pre_aligned_compositionally', 'severity': 'warning', 'message': 'Rec

LOCK EVENTS  (period-2 detector)
  total events       : 0

EITT BENCH-TEST RESIDUALS
  H_mean_full           : 1.8947679894725227
  M_sweep               : [{'M': 2, 'n_blocks': 8, 'H_mean_decimated': 1.892049224
  gate_pct              : 5.0
  note                  : Empirical observation of trajectory smoothness under tem

SANITY CHECKS  (computed during this report)
  ‖clr‖ non-finite count : 0
  ω    non-finite count : 0
  Hs   non-finite count : 0
  Σ CLR per row range   : [-2.66e-15, +1.11e-15]   (should be ≈ 0)
```

If any flag above is non-empty, the analysis on subsequent pages still ran but the reader should weight those plates accordingly. Determinism is preserved either way.

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Plate 3 | Evolution of Proportions (raw + cumulative)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

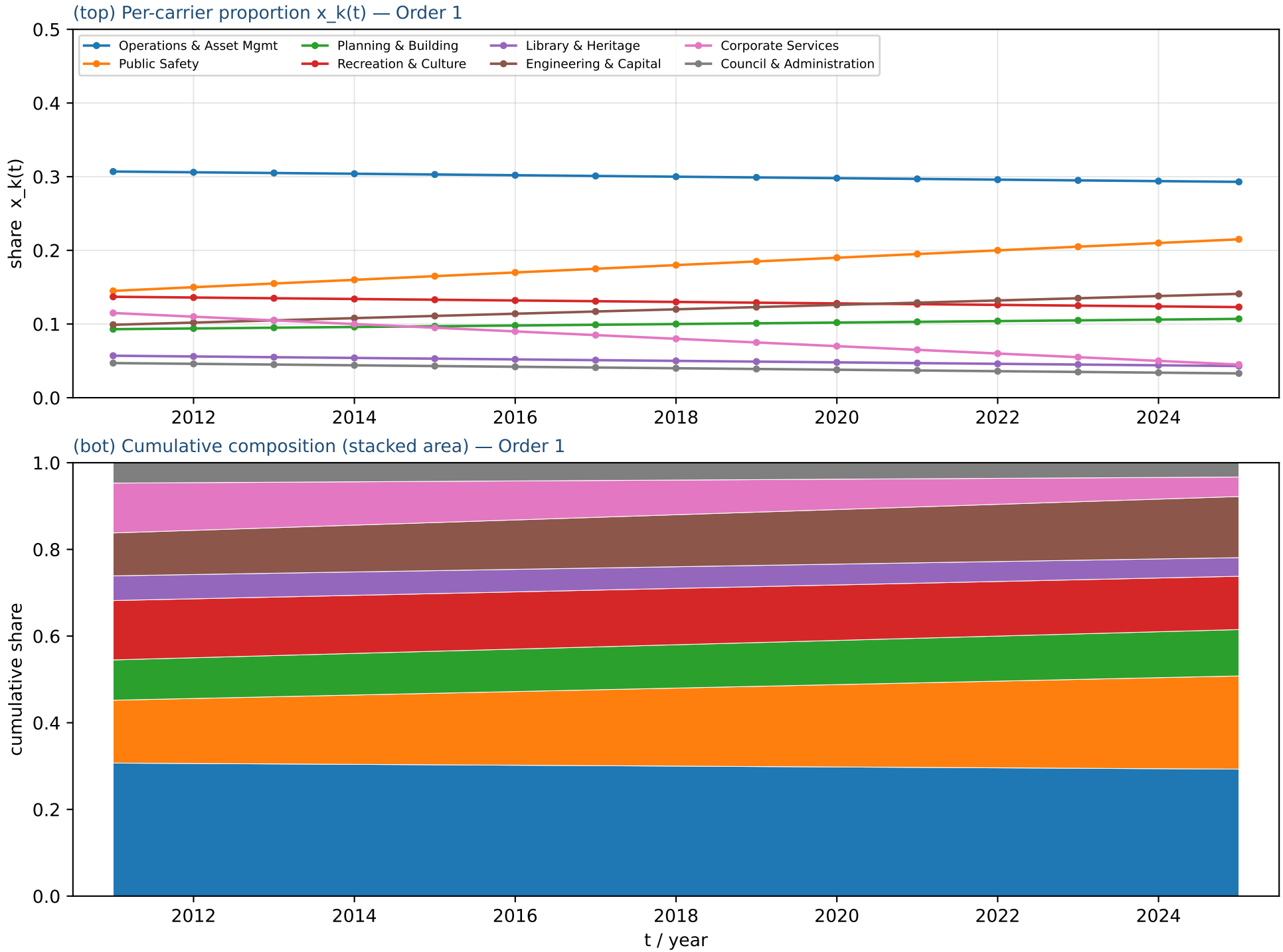

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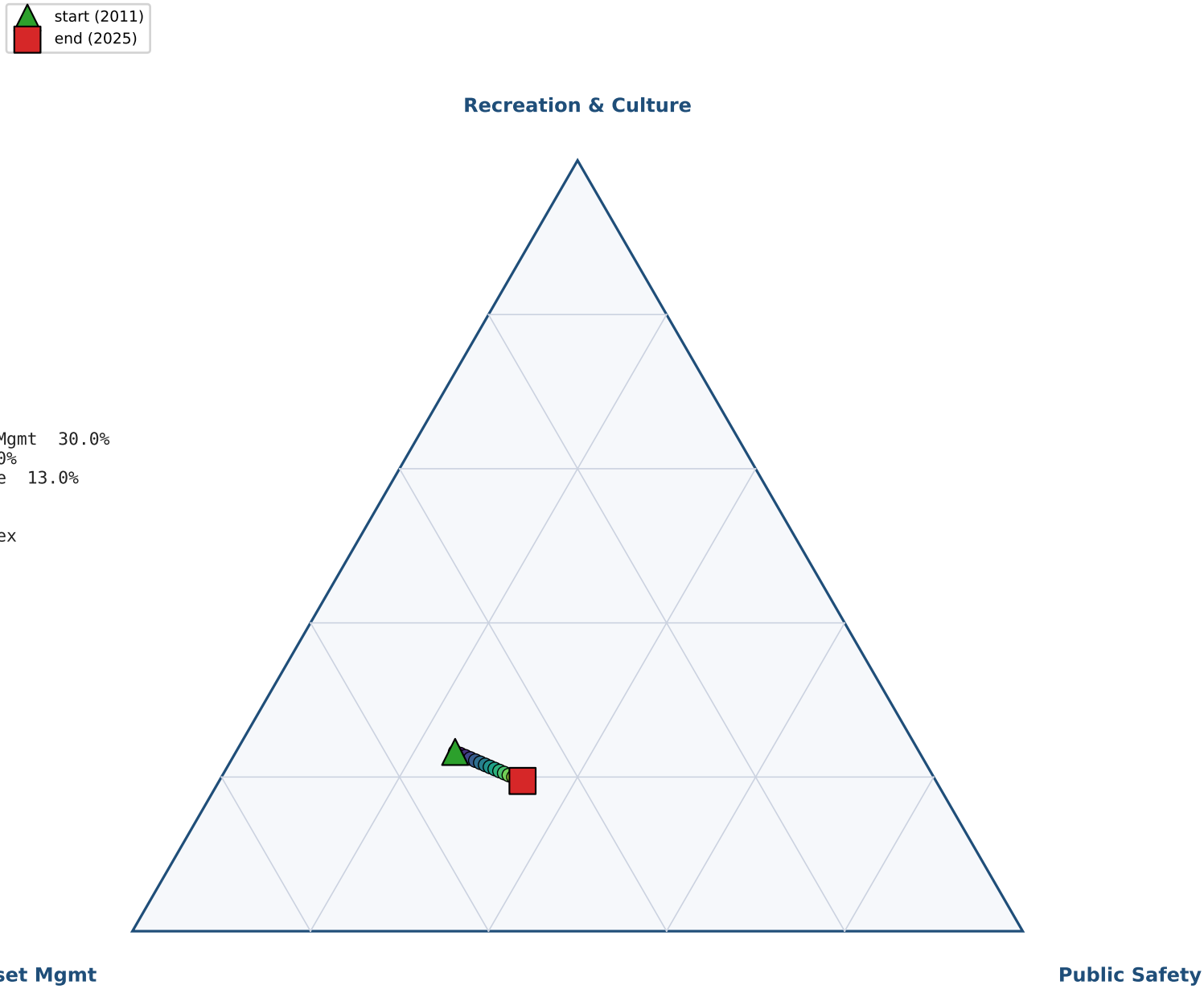
Plate 4 | Ternary Diagram (top-3 carriers)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

3-part subcomposition closed over Operations & Asset Mgmt, Public Safety, Recreation & Culture
(remaining 5 carriers absorbed into closure)




```
Top-3 by mean share:
  Operations & Asset Mgmt  30.0%
  Public Safety   18.0%
  Recreation & Culture  13.0%

T = 15 timesteps
colour map = time index
▲ start  ■ end
```

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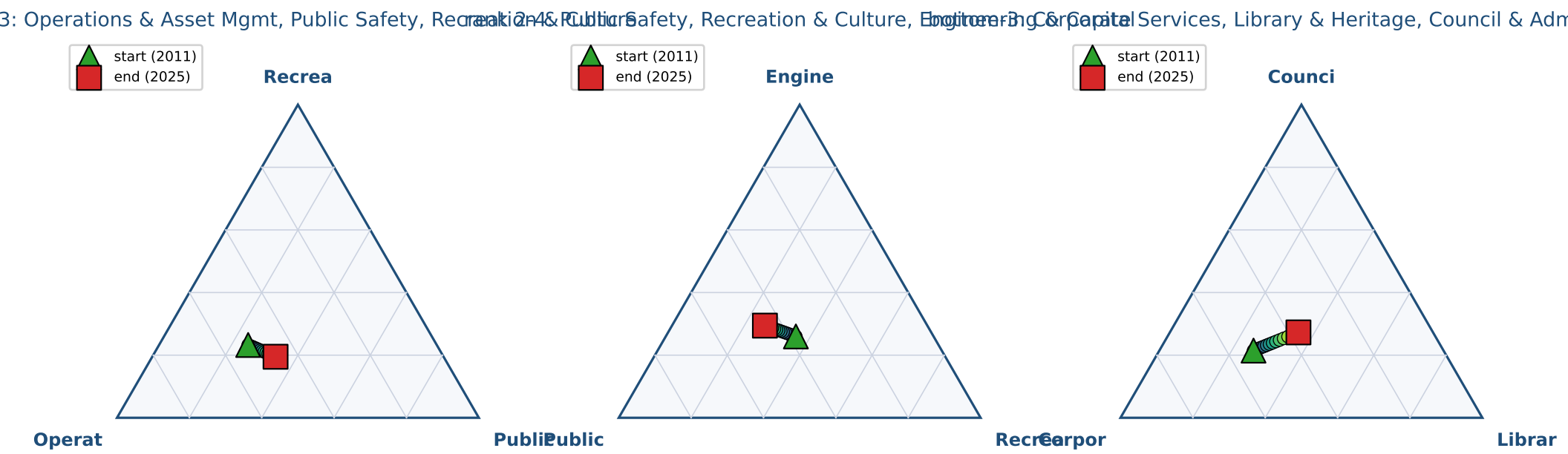
Plate 5 | Three-Part Subcompositions (3 ternaries)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Each ternary closes the chosen 3 carriers; remaining carriers absorbed into closure. ▲ start ■ end | colour map = time index

Triplet source: default heuristic

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Plate 6 | Variation matrix τ_ij heatmap

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

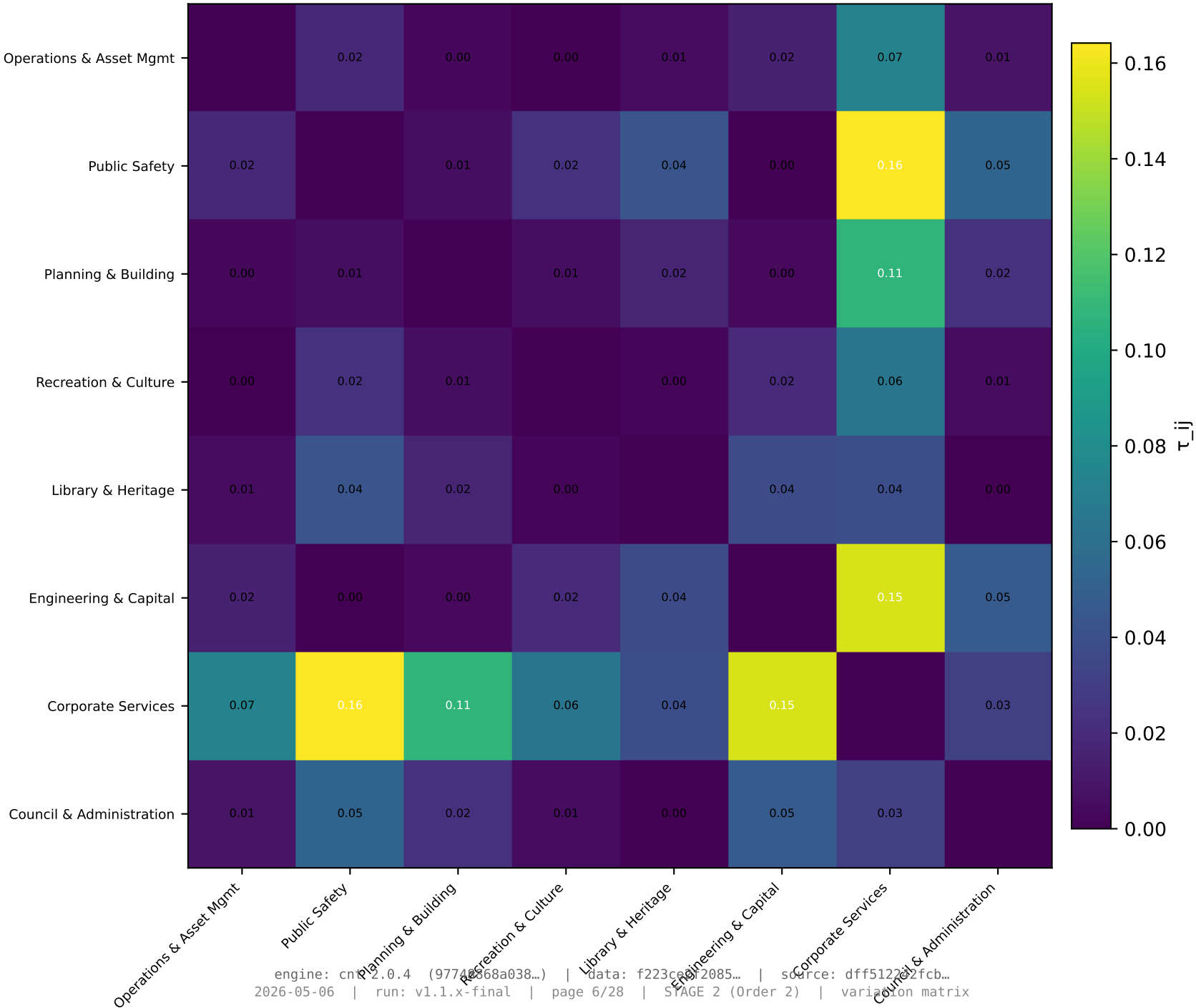

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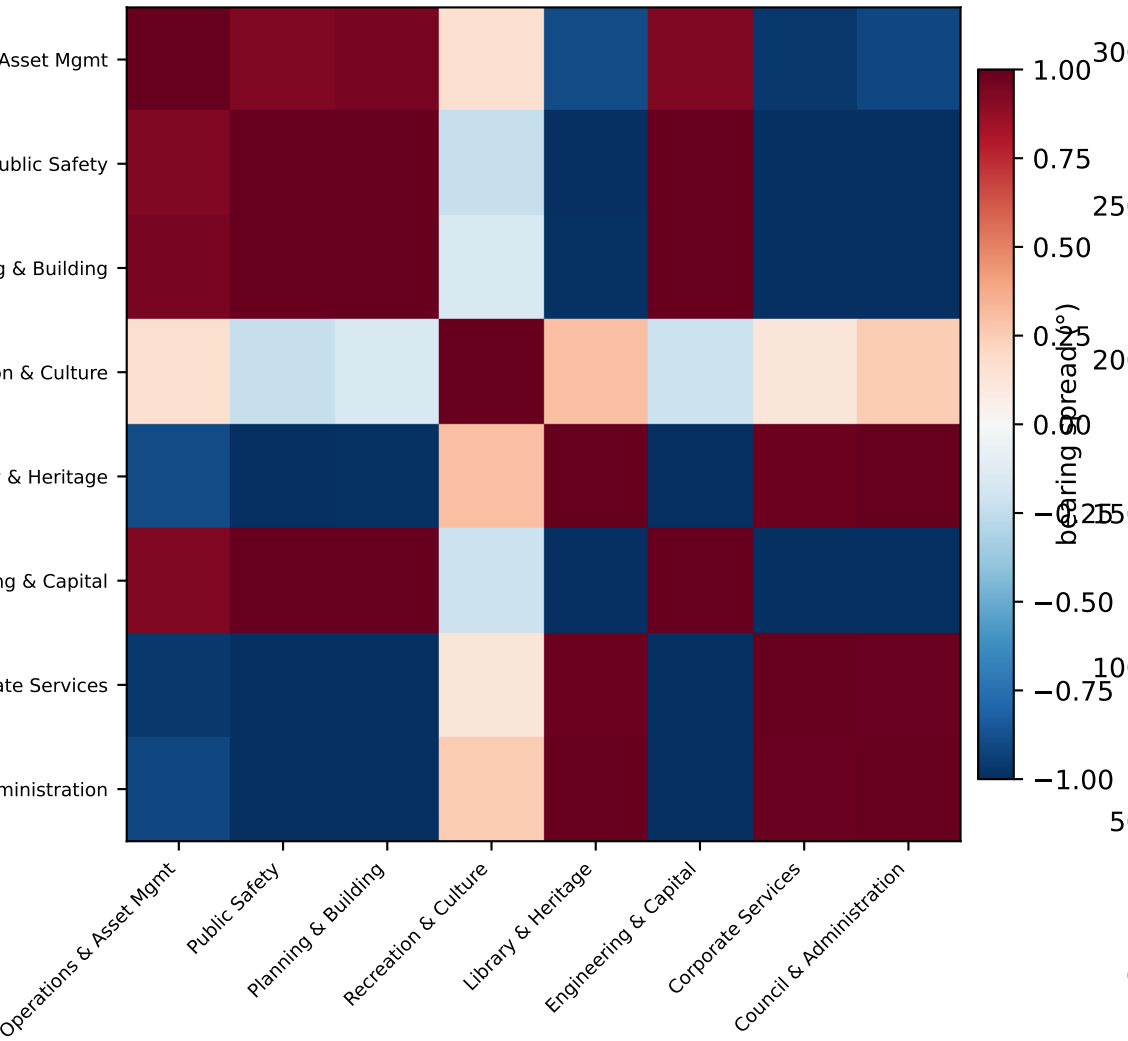
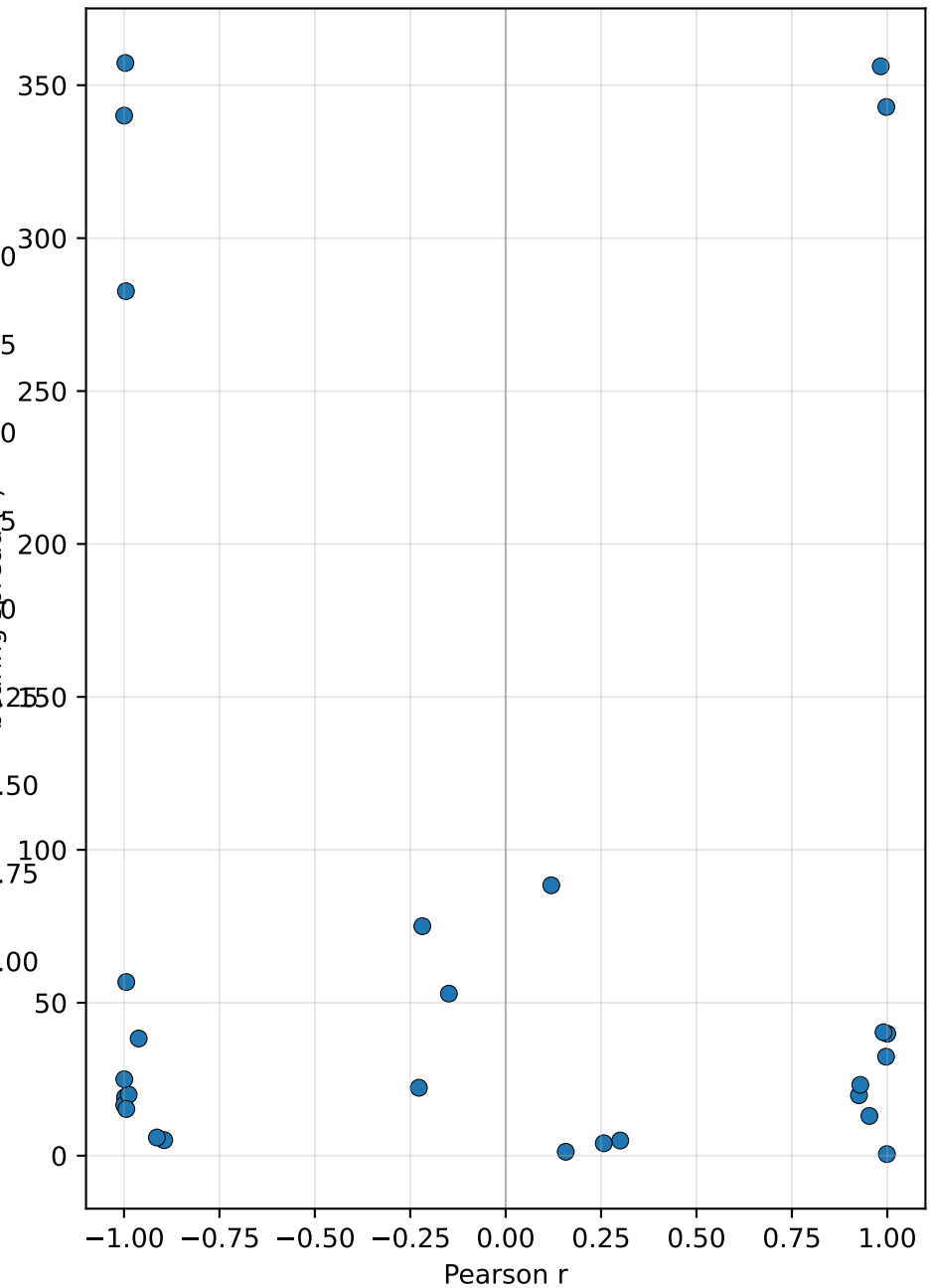
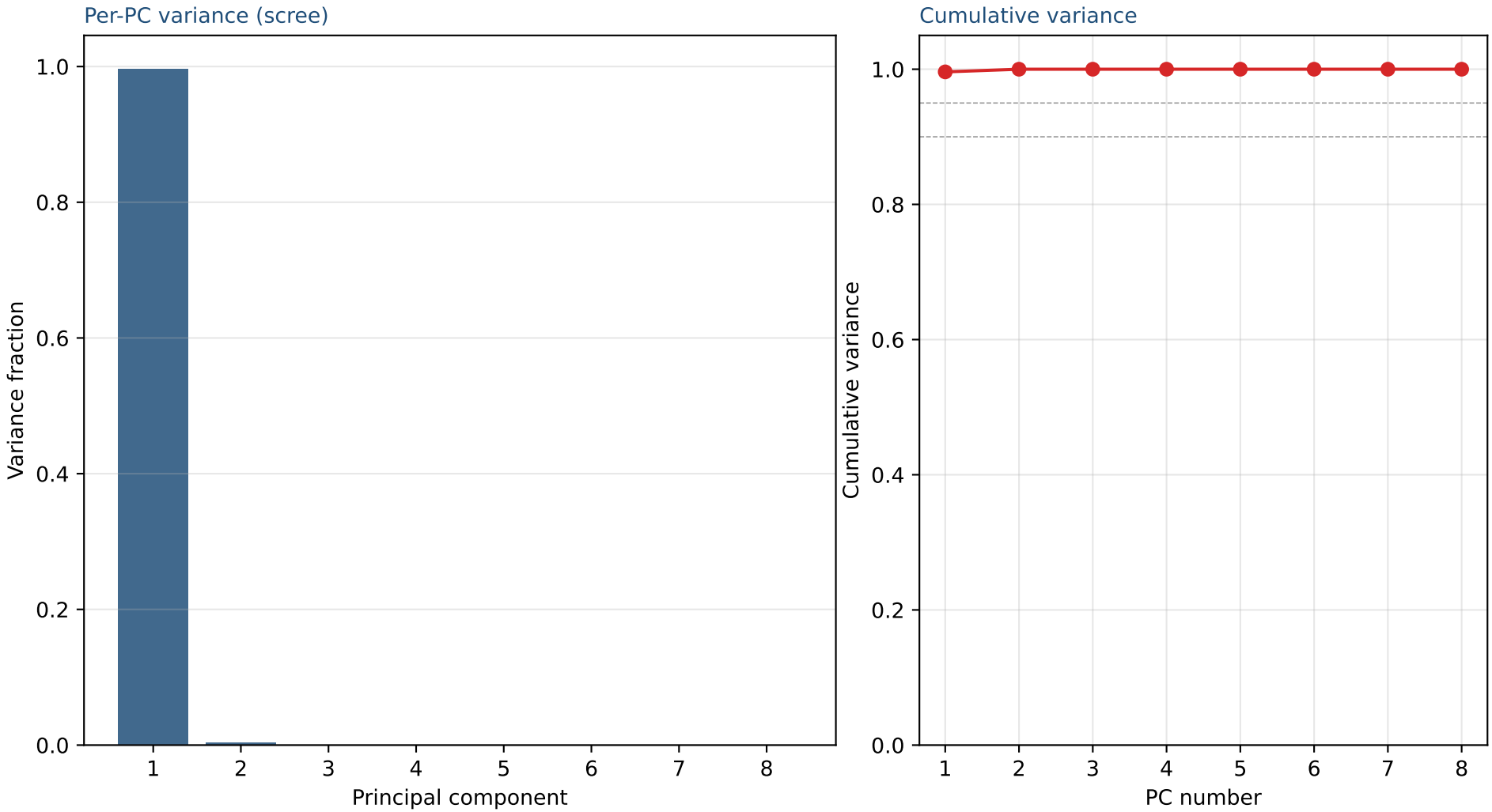
Plate 7 | Pair Pearson r heatmap + co-movement scatter

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2








```
Per-PC variance contribution:
  PC1: 99.60%   cum 99.60%
  PC2:  0.39%   cum 100.00%
  PC3:  0.00%   cum 100.00%
  PC4:  0.00%   cum 100.00%
  PC5:  0.00%   cum 100.00%
  PC6:  0.00%   cum 100.00%
  ... 2 more PCs
```

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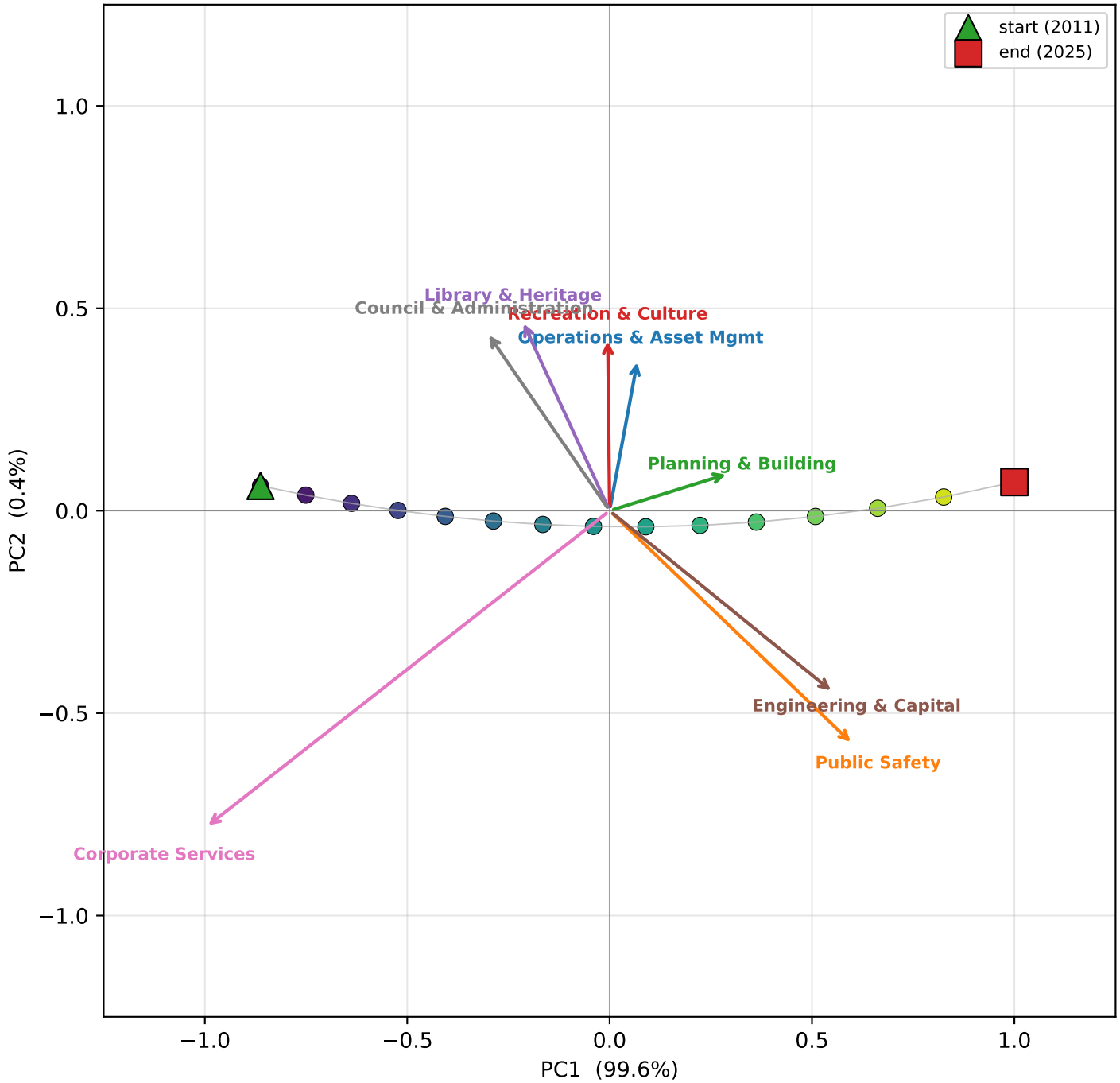
Plate 9 | CLR Biplot (CoDa biplot)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

CoDa biplot - covariance form, 2D variance explained 100.0%




```
CoDa biplot (Aitchison-Greenacre):
  - Points: timestep scores in PC(1,2) of c
  - Rays:   carrier loadings (columns of V)
  - Ray length = carrier importance on the
  - Angle between rays ~ negative log-ratio
  - Distance between point projections = ap
    of the Aitchison distance restricted to

PC1 captures 99.6% of variance.
PC2 captures 0.4% of variance.
Total 2D plane:  100.0%
```

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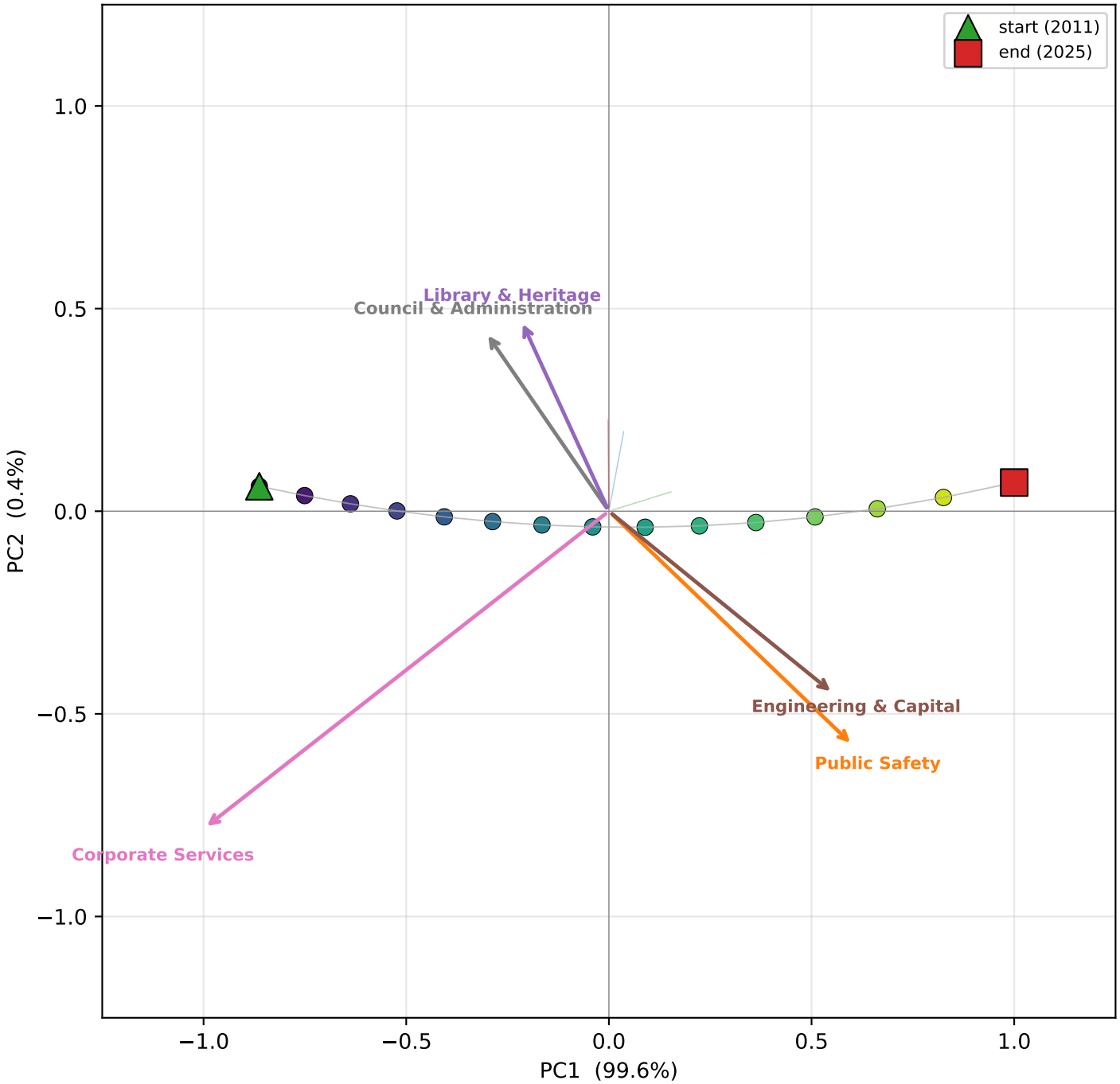
Plate 10 | CLR Biplot — top-5 carriers by loading magnitude

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

Top-5 biplot — 2D variance 100.0% (D = 8, dropped 3 short rays)



Loading magnitudes $\sqrt{(G_1^2 + G_2^2)}$:

- Corporate Services 0.9206
- Public Safety 0.6056
- Engineering & Capital 0.5180
- Council & Administration 0.3888
- Library & Heritage 0.3761

Dropped (faint stubs only):
Recreation & Culture 0.3112
Operations & Asset Mgmt 0.2775
Planning & Building 0.2257

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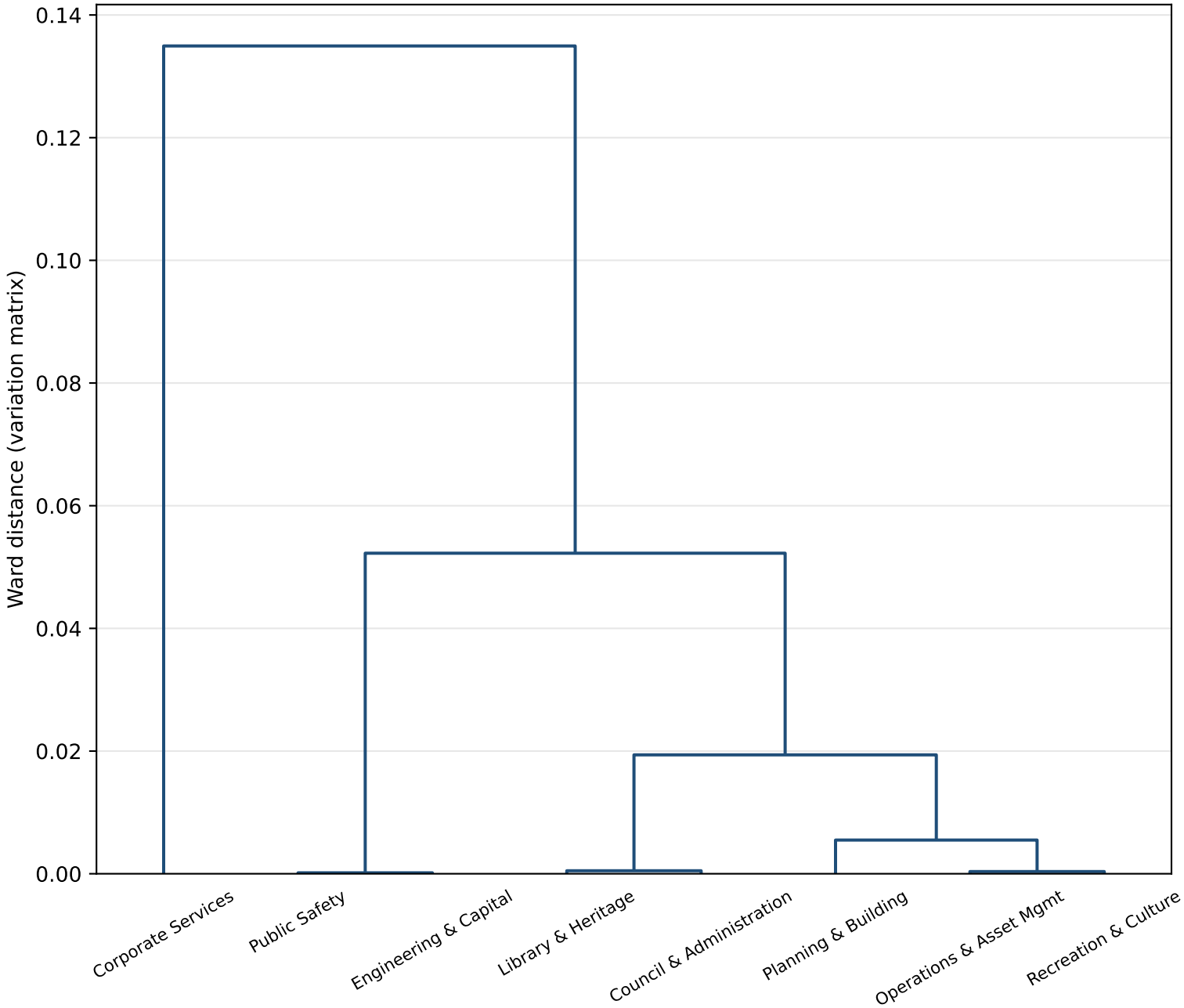
Plate 11 | Balance-Dendrogram (Ward on variation matrix)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

Hierarchical balance tree — each merge defines one balance partition




```
Balance-dendrogram (Pawlowsky-Glahn-
  – Ward clustering on the (symmetri
  – Each internal node merges two ca
  – That merge defines a balance:
      b = sqrt(|A||B|/(|A|+|B|)) ·
    where g() is the geometric mean.
  – The balances form an orthonormal

D = 8 carriers → D-1 = 7 balances.
Tightly-clustered carriers (low vari
See companion plate for the full SBP
```

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Plate 12 | Sequential Binary Partition (SBP) Matrix

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

SBP cells: + = numerator, - = denominator, . = not in this balance

	b1	b2	b3	b4	b5	b6	b7
t Mgmt	.	+	.	-	-	-	-
Safety	+	.	.	.	.	+	-
ilding	.	.	.	+	-	-	-
ulture	.	-	.	-	-	-	-
ritage	.	.	+	.	+	-	-
apital	-	.	.	.	.	+	-
rvices	.	.	.	.	.	.	+
ration	.	.	-	.	+	-	-

Balance b_k = sqrt(|A_k||B_k|/(|A_k|+|B_k|)) * ln(g(A_k) / g(B_k)) | D = 8; D-1 = 7 balances.

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Plate 13 | Hs (Higgins scale) trajectory — Order 2 aggregate

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

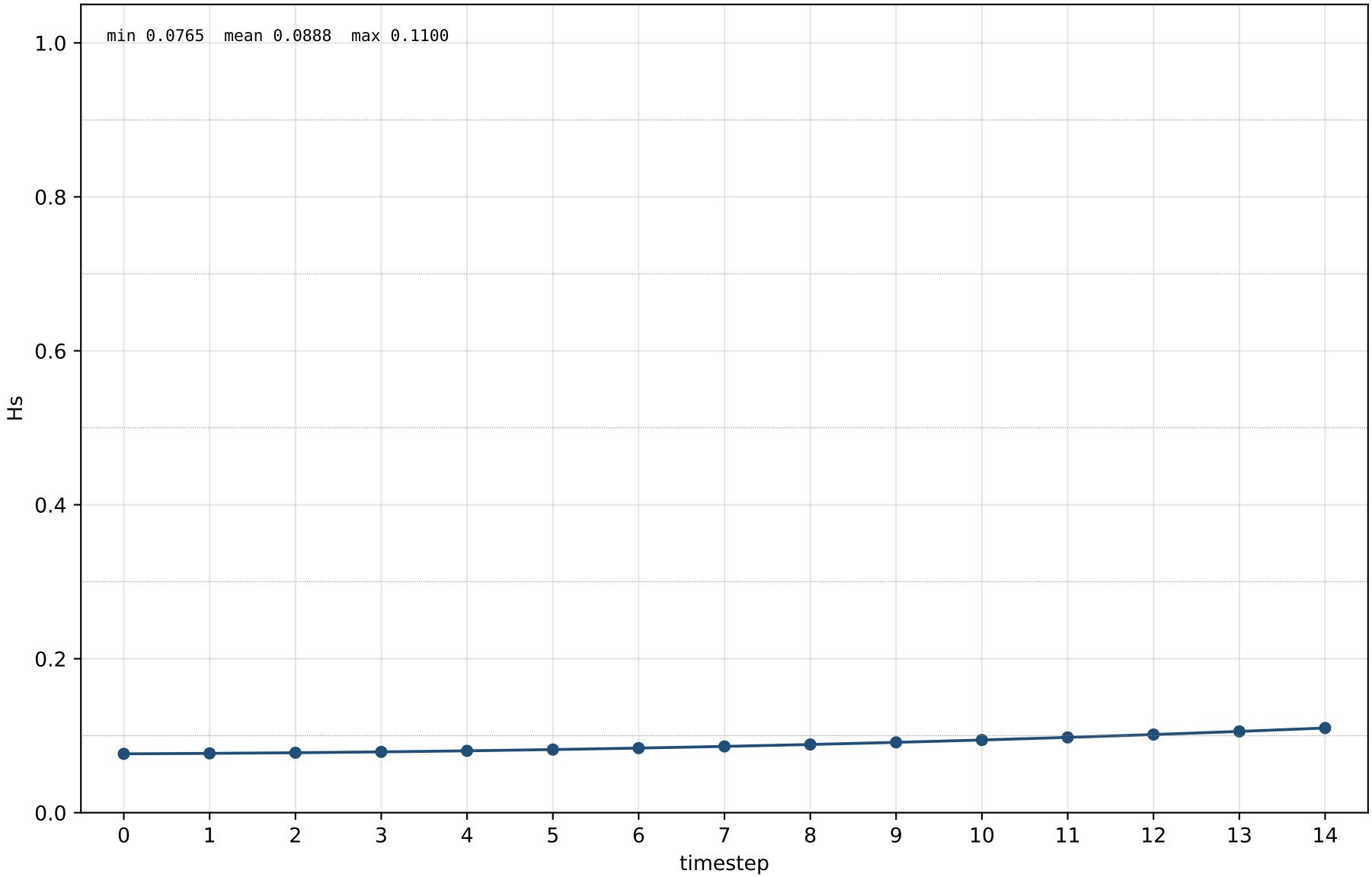

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Plate 14 | Aitchison norm trajectory ‖clr(x)‖₂ — Order 2 aggregate

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

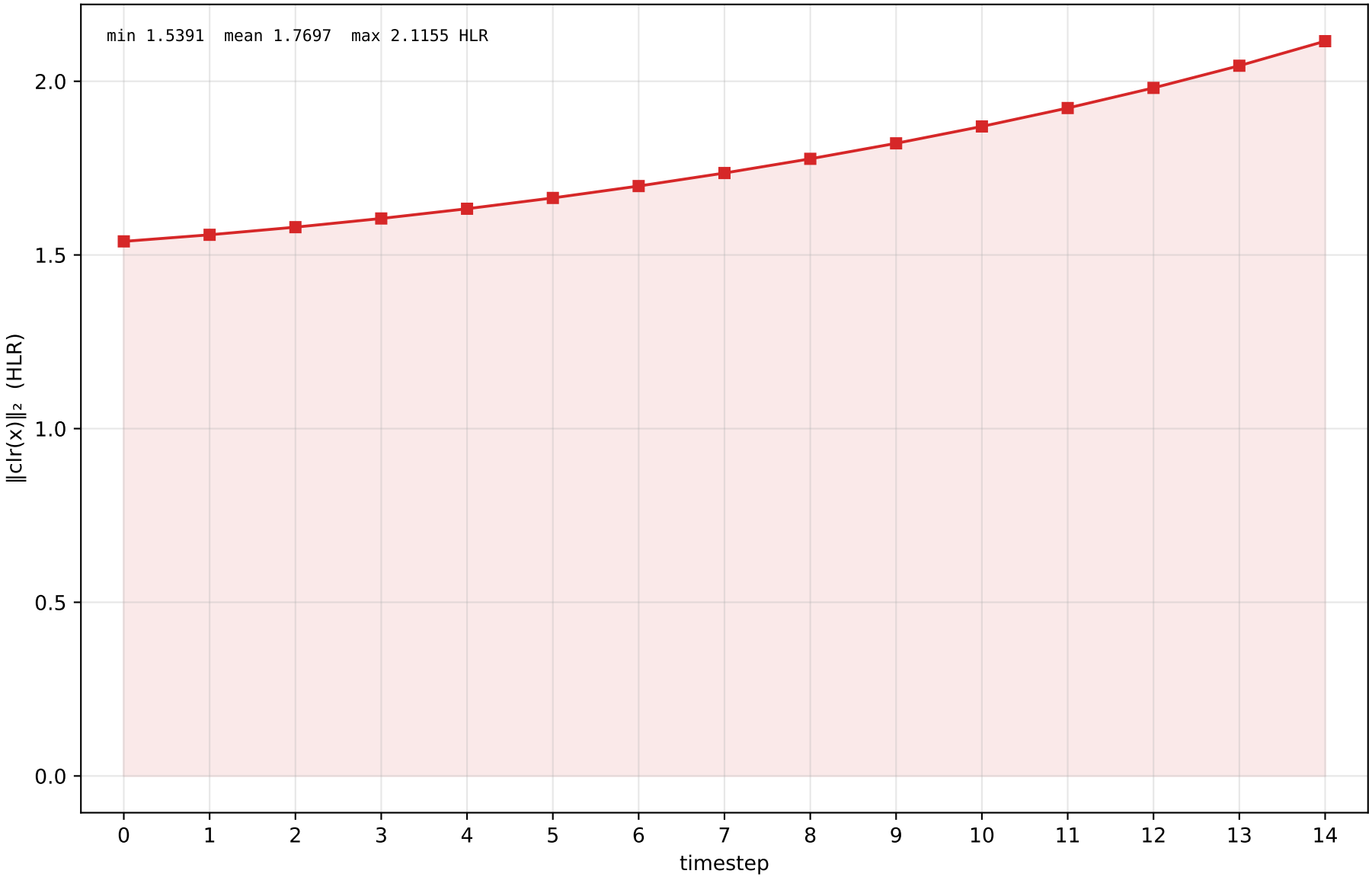

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Plate 15 | ILR axes 1–3 trajectories (Helmert basis) — Order 2 aggregate

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

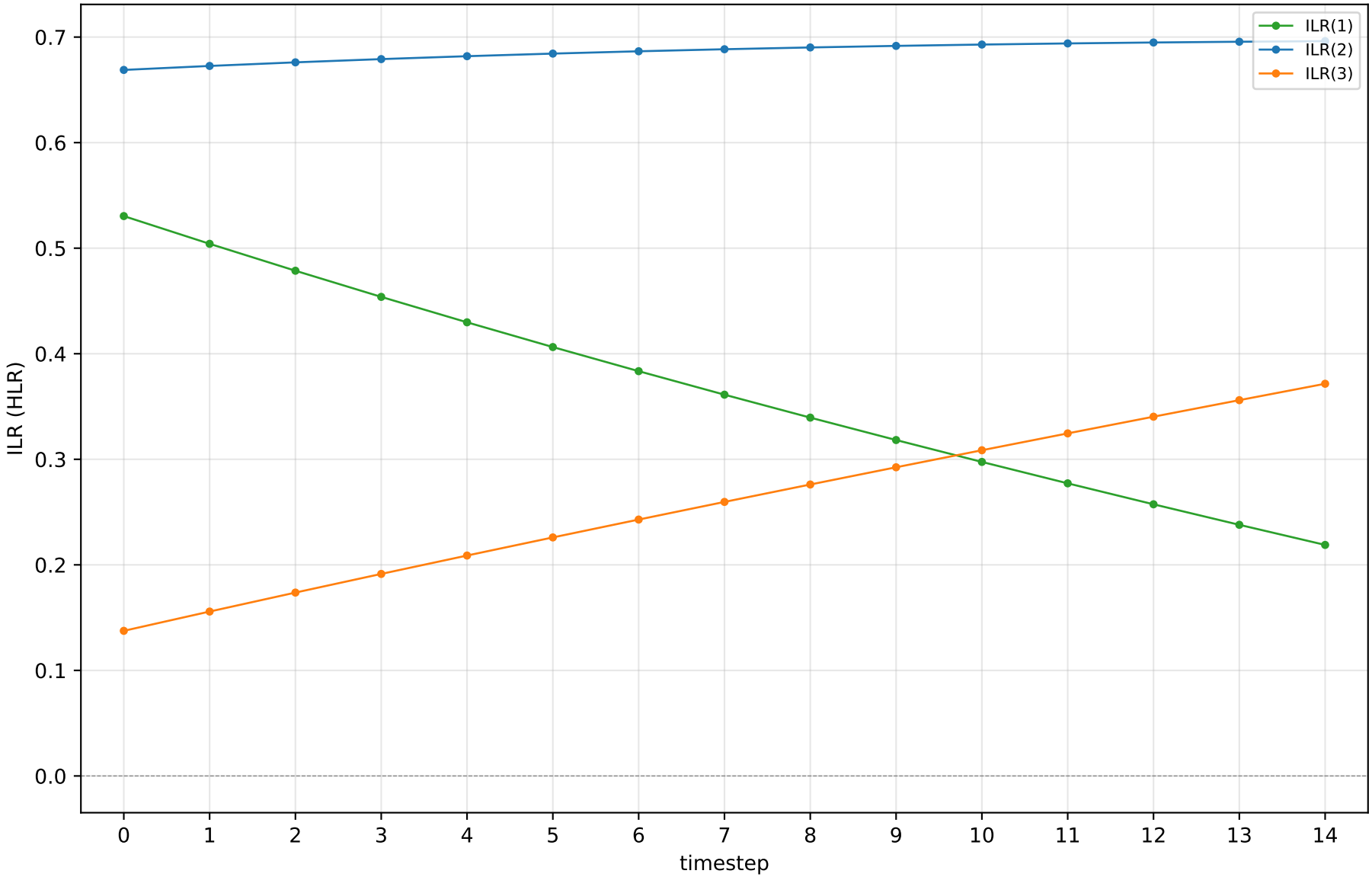

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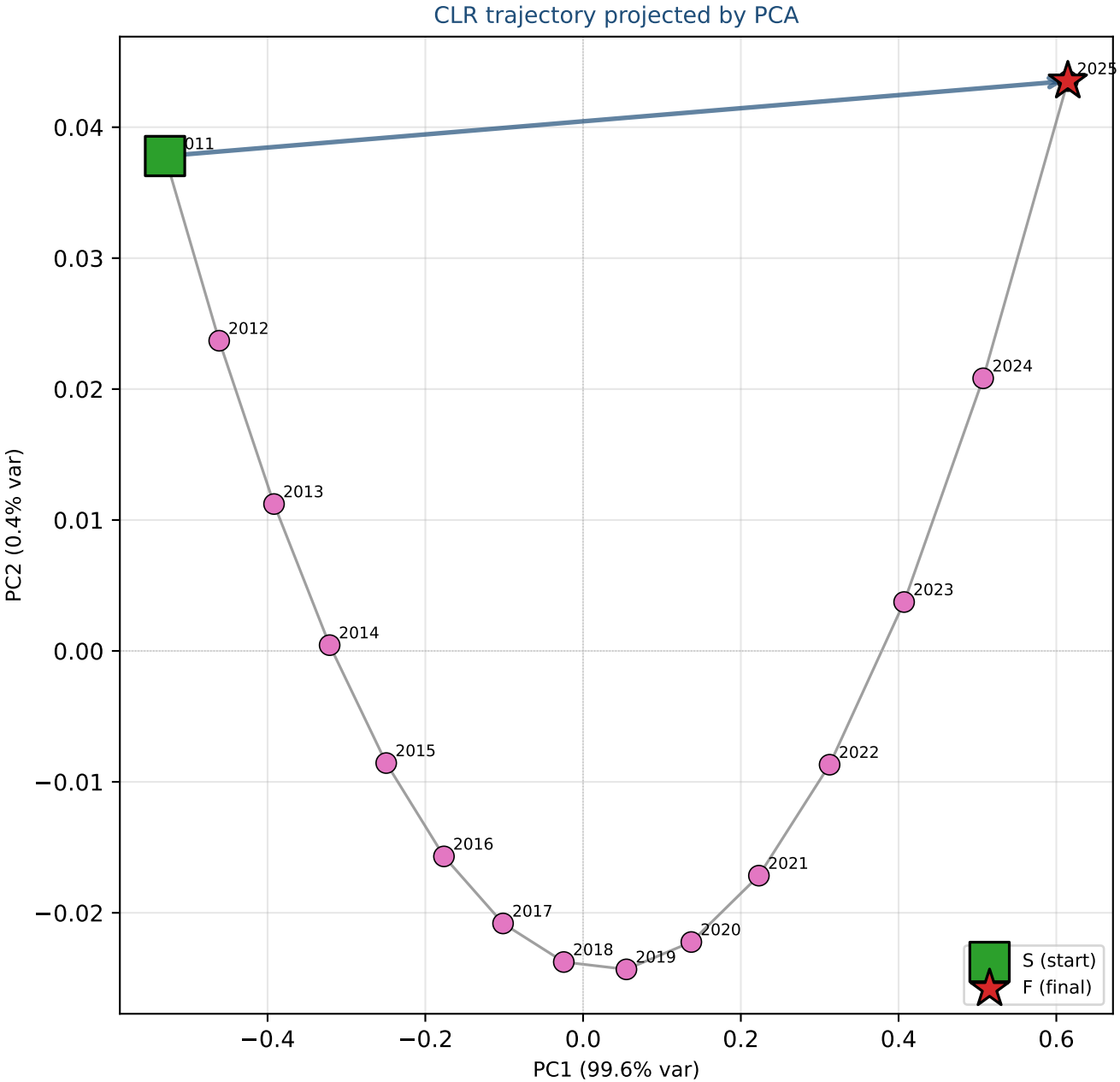
Plate 16 | System Course Plot — PCA 2D path with S→F net vector

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Navigation metrics (V_net = h_F − h_S)

```
h_start label    : 2011
h_final label    : 2025

net_distance     : 1.1447 HLR
path_length      : 1.1544 HLR
course_directness: 0.9916
  (1.0 = straight line, 0 = pure looping)

dynamic range S  : 1.8767 HLR
dynamic range F  : 2.1837 HLR

Reading guide:
  tight cluster  → stable composition
  long excursion → structural shift
  sharp turn     → regime transition
  loop           → cyclic behaviour
  spiral         → systematic drift
```

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Plate 17 | Polar bearing rose (4 sample timesteps)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

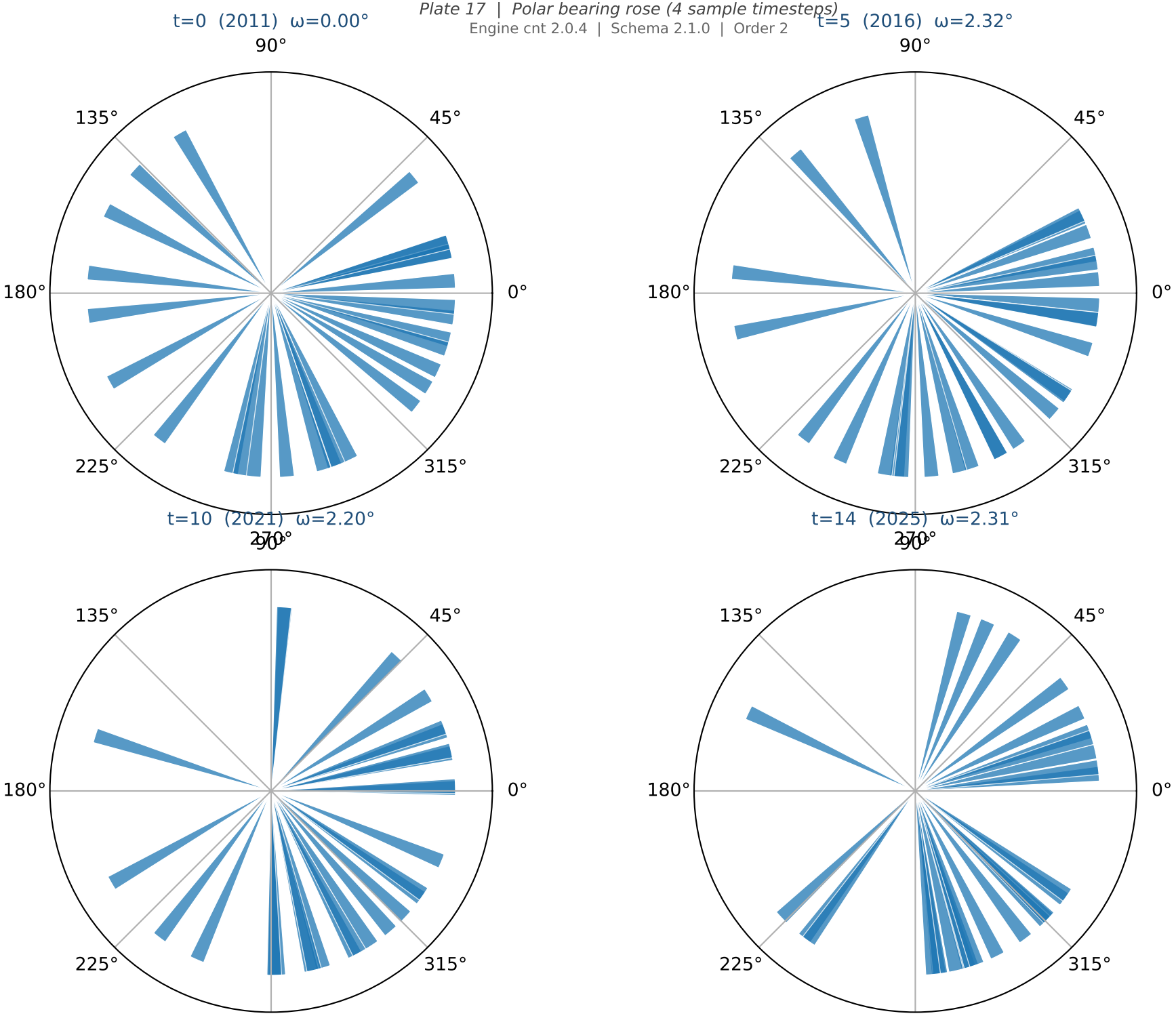

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Plate 18 | Bearing heatmap T × C(D,2) pairs

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

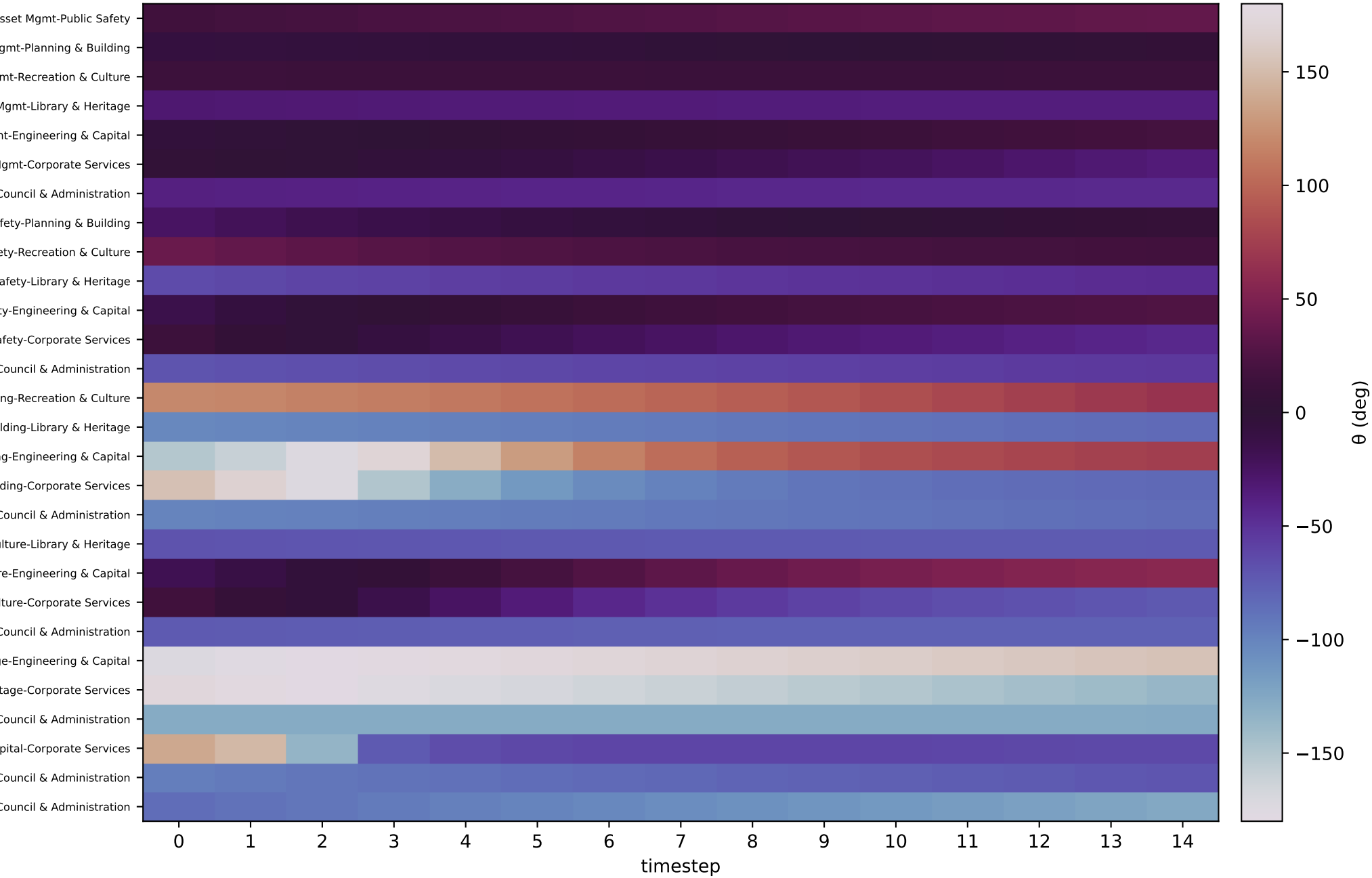

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Plate 19 | ω angular velocity trajectory

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

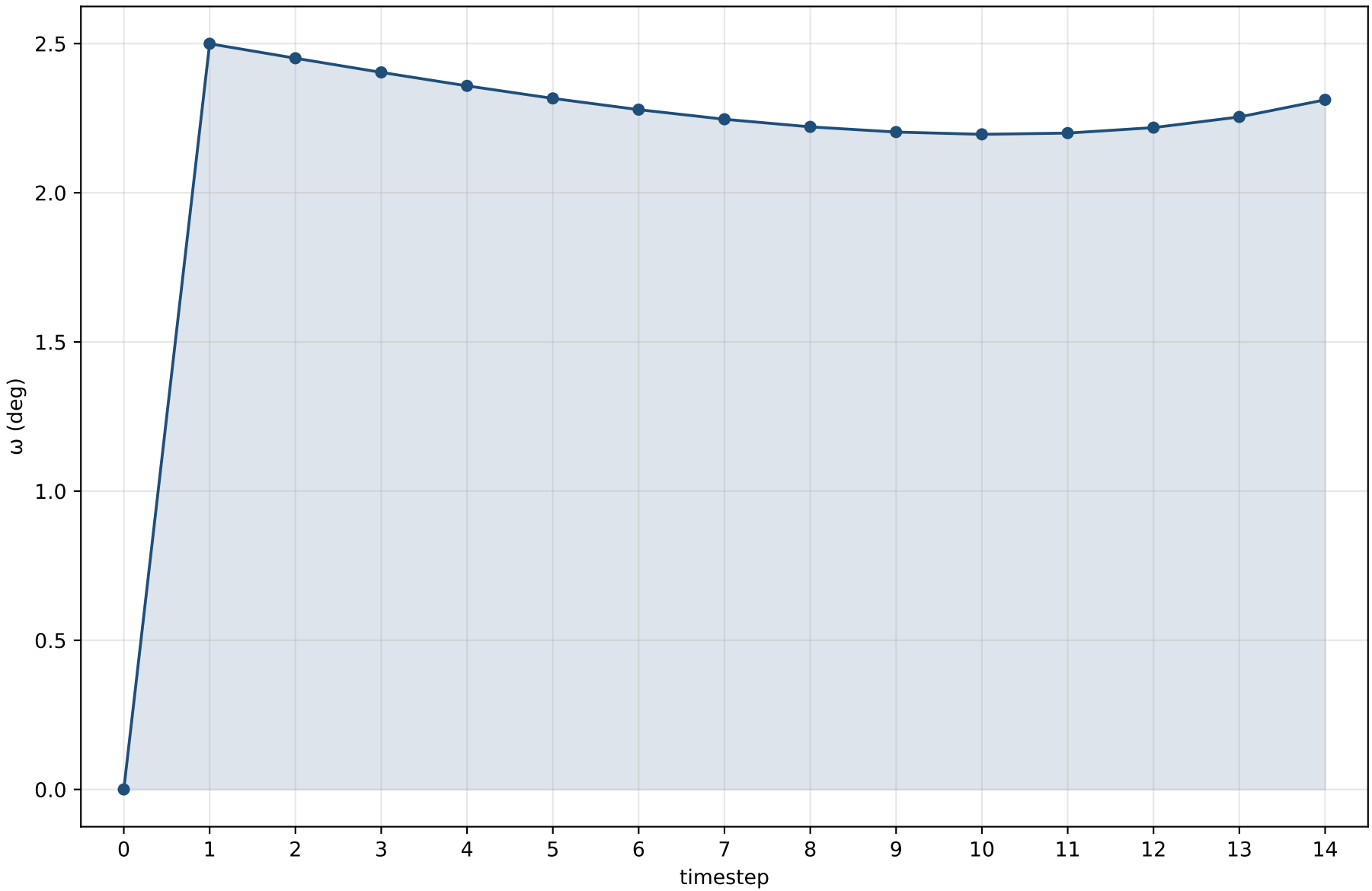

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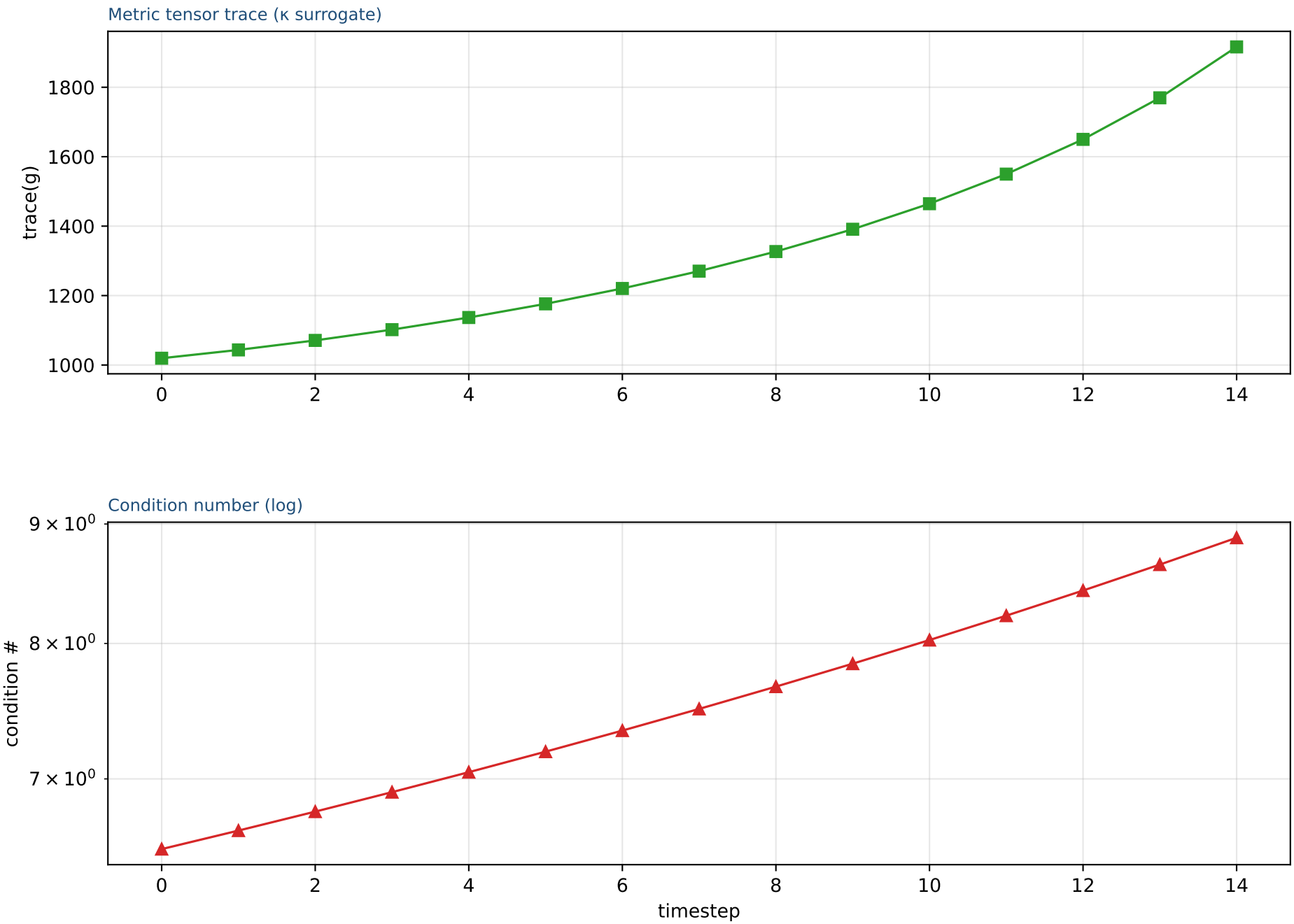
Plate 20 | Metric tensor trace + condition number

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, Markham municipal operating budget 2011-2025

Plate 21 | Helmsman σ lineage timeline

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

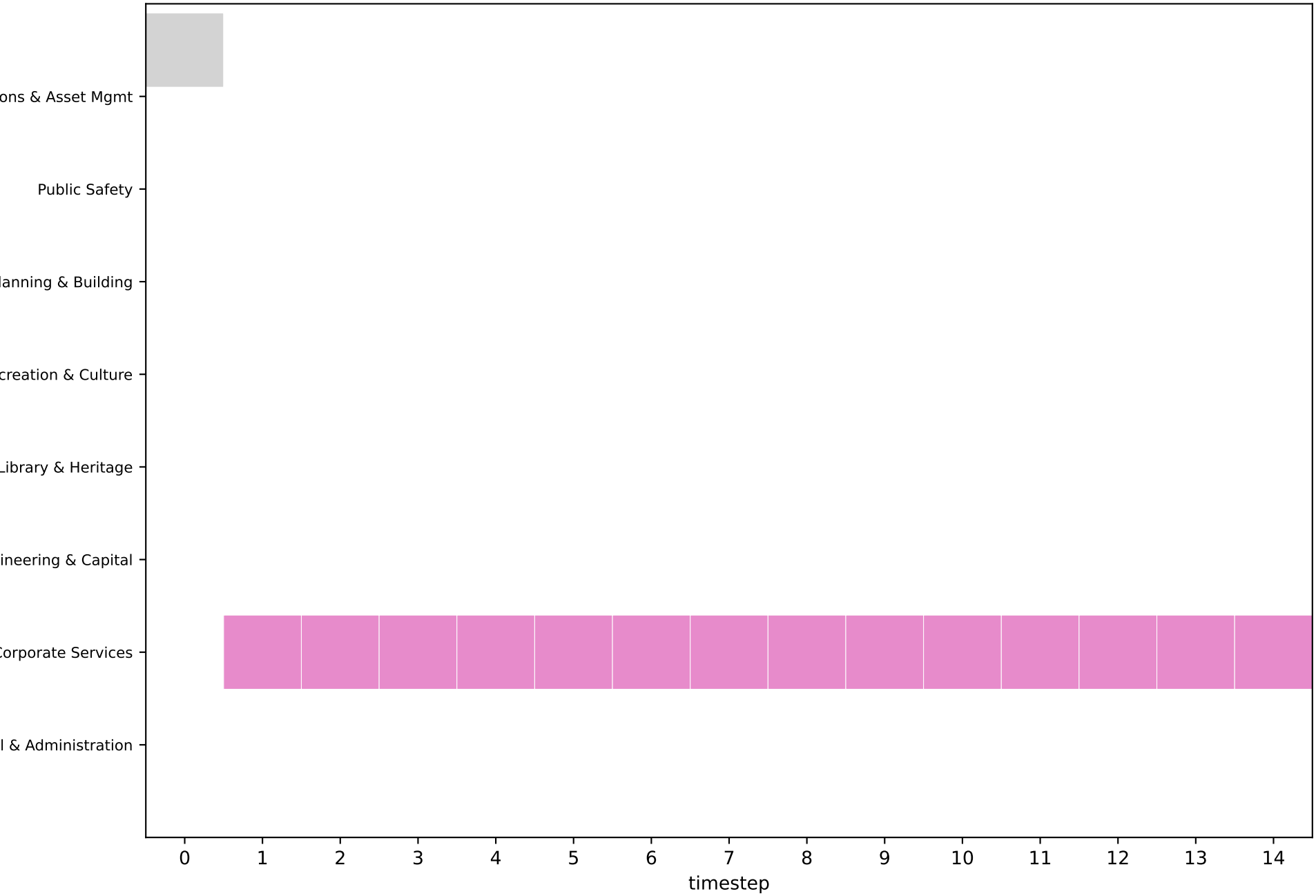


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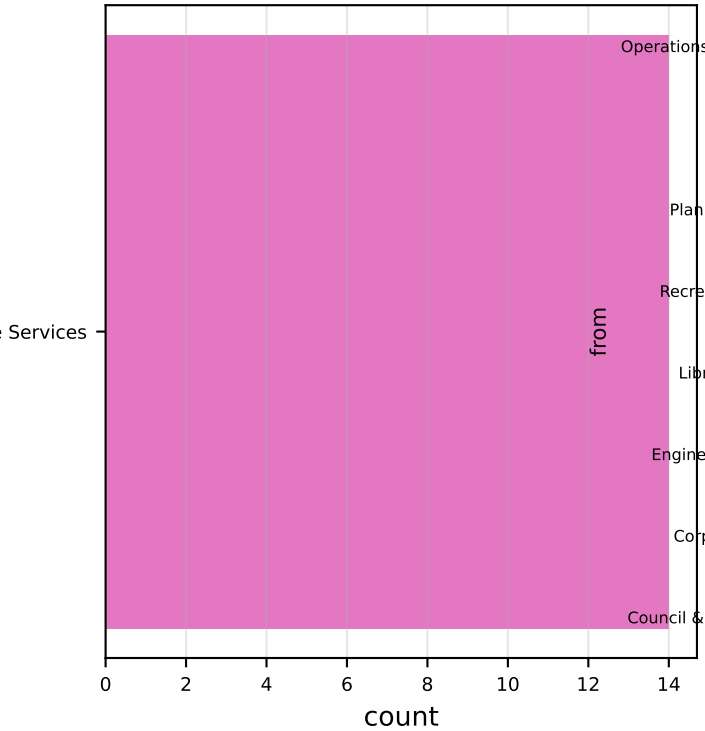
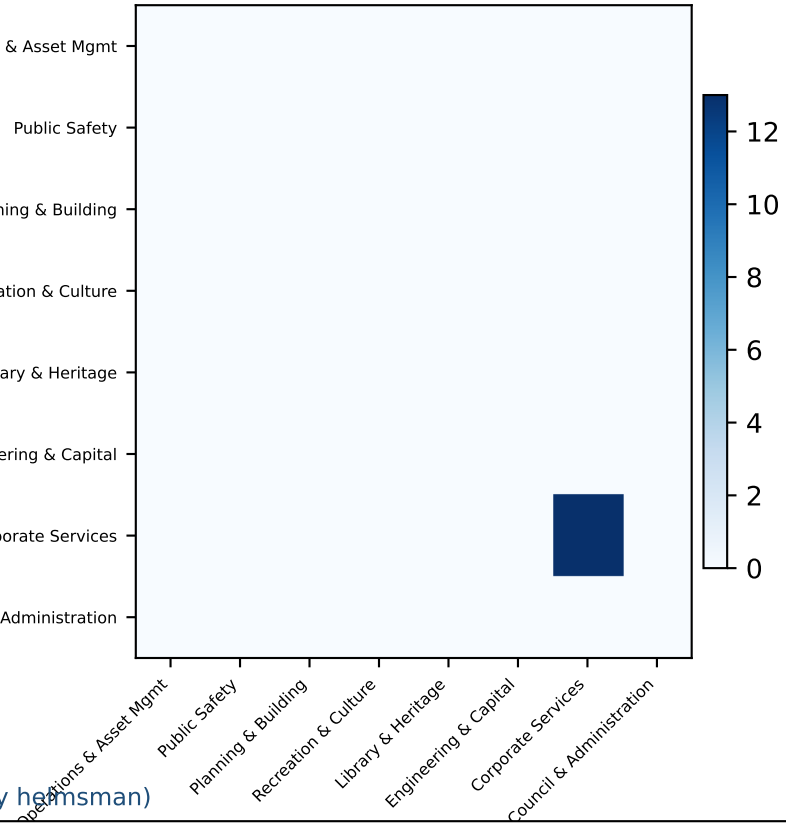
Plate 22 | Helmsman σ — frequency · transitions · per-year displacement

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

Helmsman frequency



Transition matrix



Helmsman displacement per year (bar coloured by helmsman)

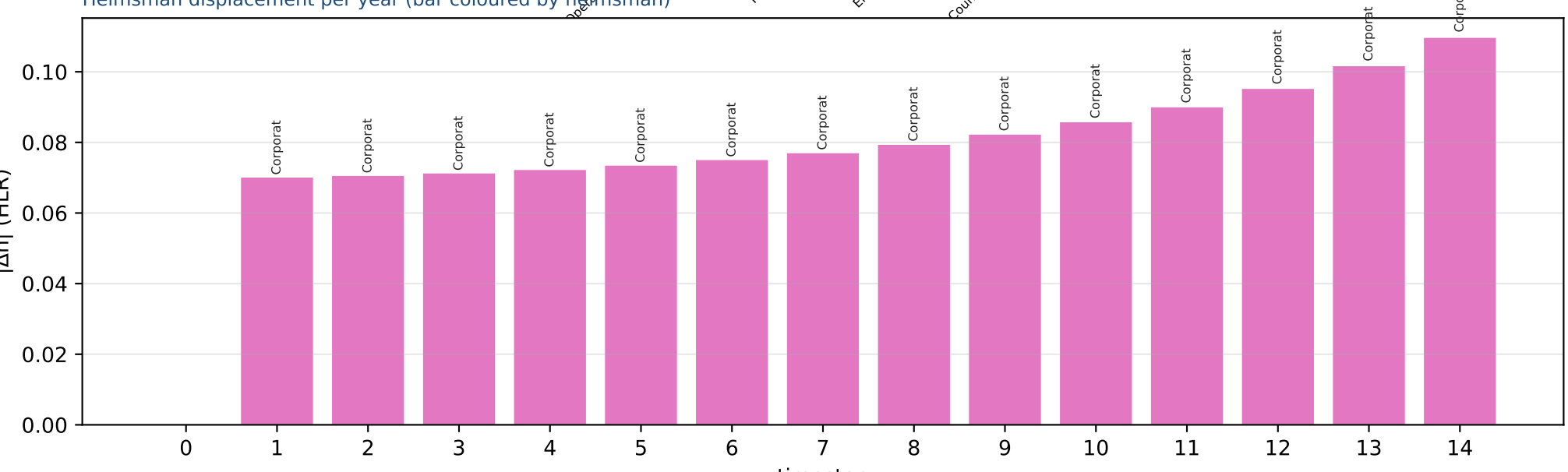


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Plate 23 | Ring class evolution + Hs

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

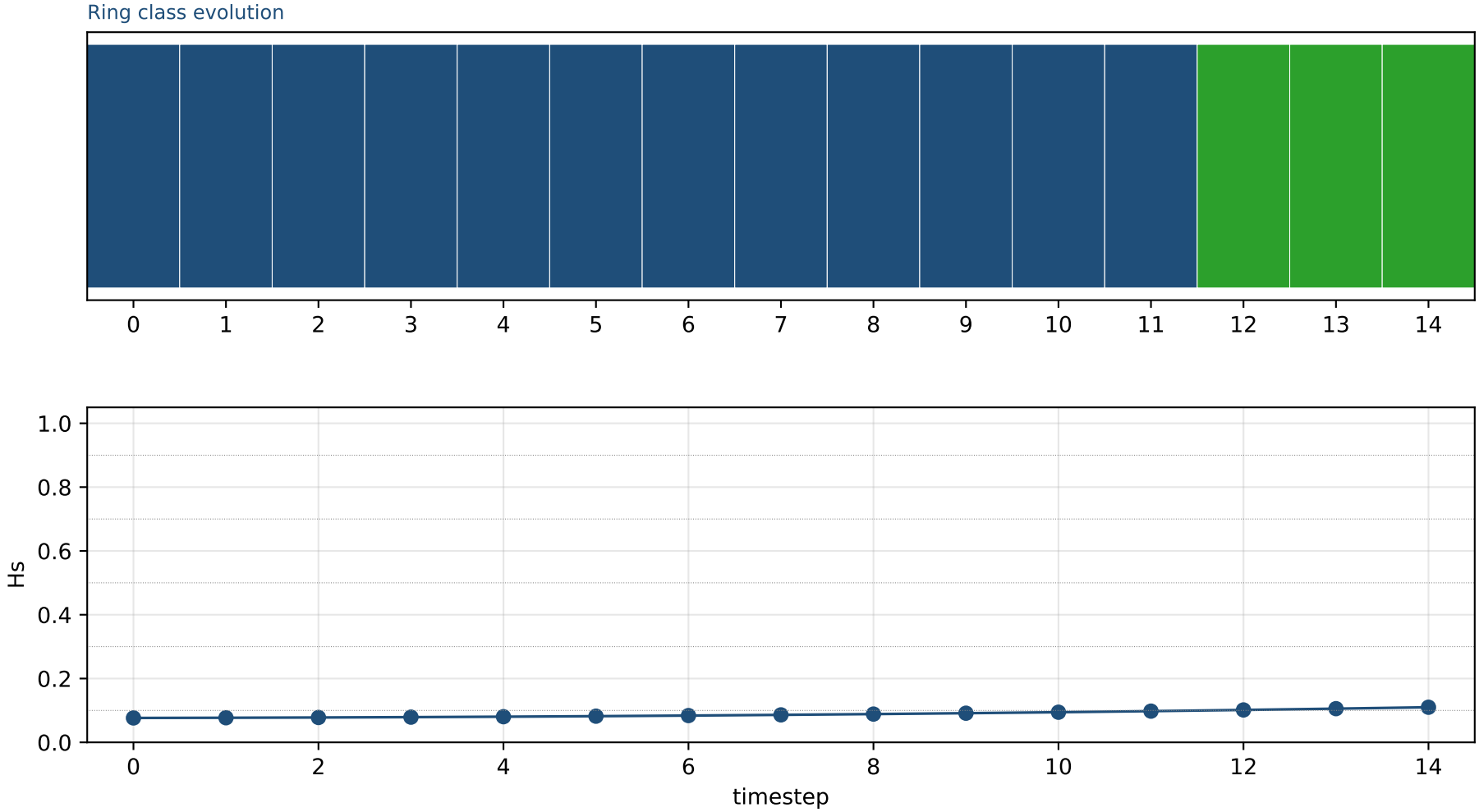

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Plate 24 | Lock event timeline (carriers × timesteps)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

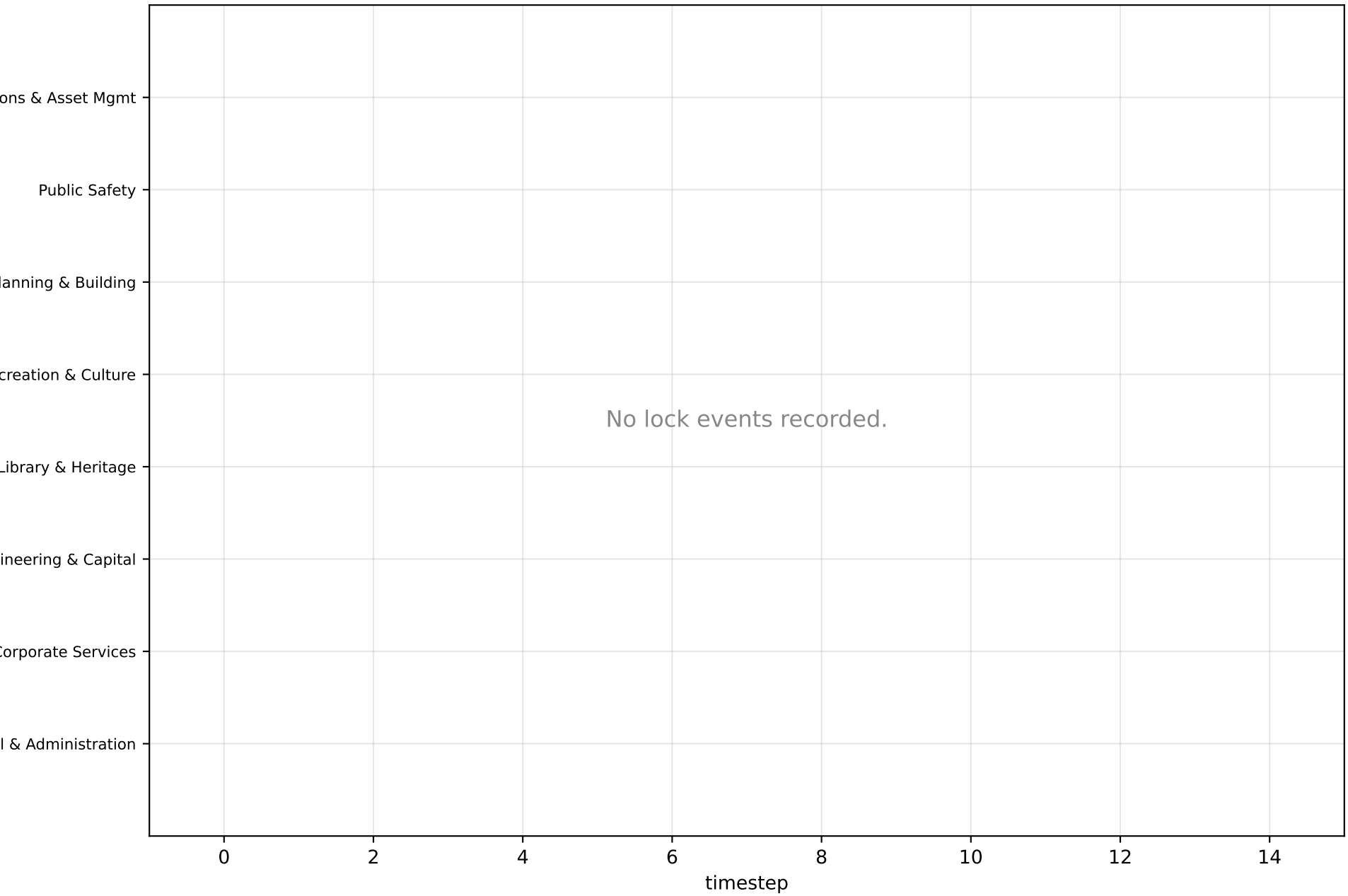

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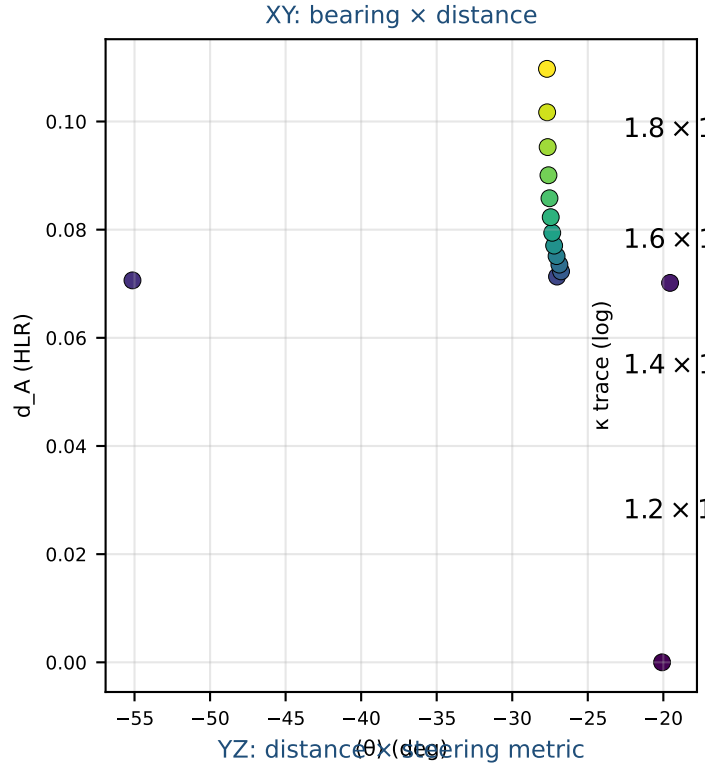
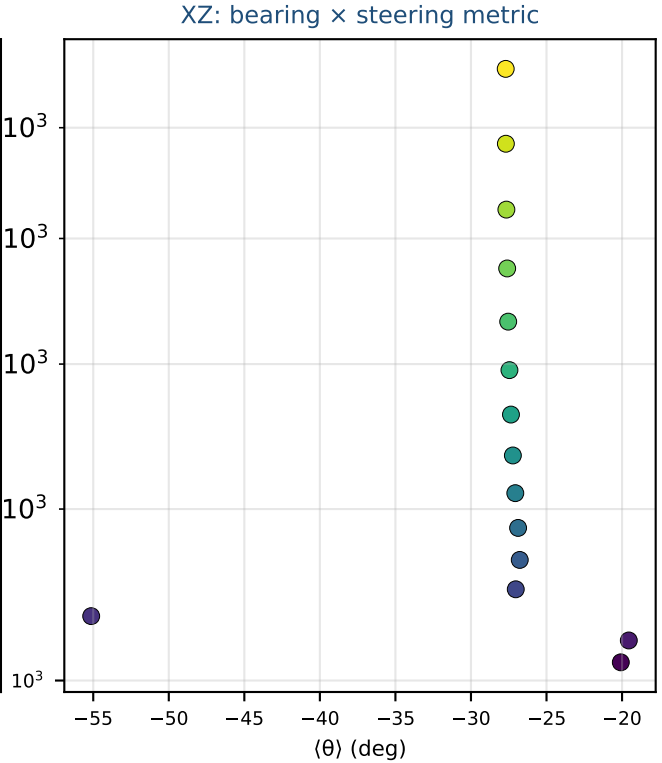
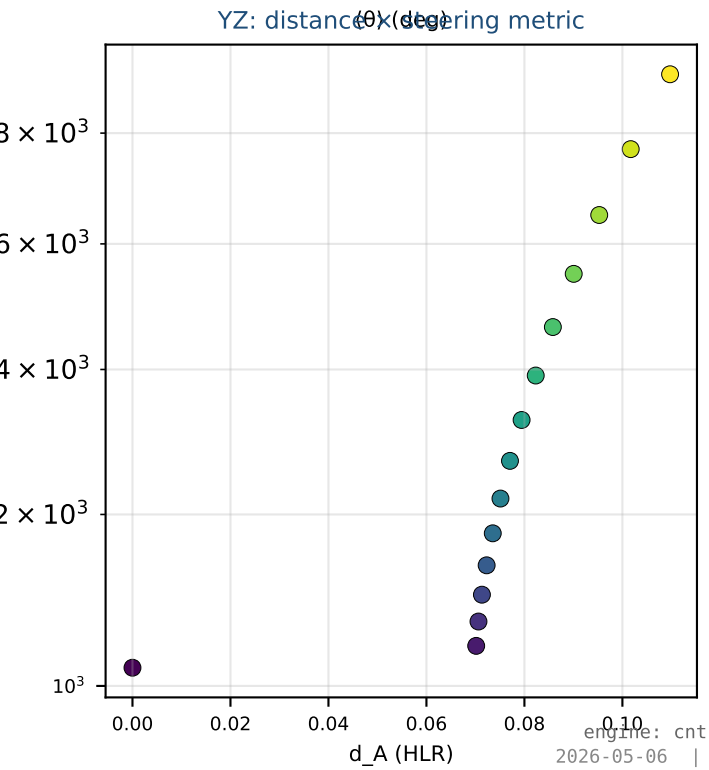
Plate 25 | Compositional Bearing Scope (CBS) — 3 orthogonal faces (Math Handbook Ch 16)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2







CBS info panel (Math Handbook Ch 16)

```
⟨θ⟩ range  : [-55.1, -19.6]°
d_A range  : [0.0000, 0.1097] HLR
κ range    : [1.02e+03, 1.92e+03]
Hs range   : [0.0765, 0.1100]
ω max      : 2.50°

Each face: one observation per timestep.
Colour = timestep (early=dark, late=bright).
Three faces locate the system in the
(bearing, distance, steering) measurement cube.
```

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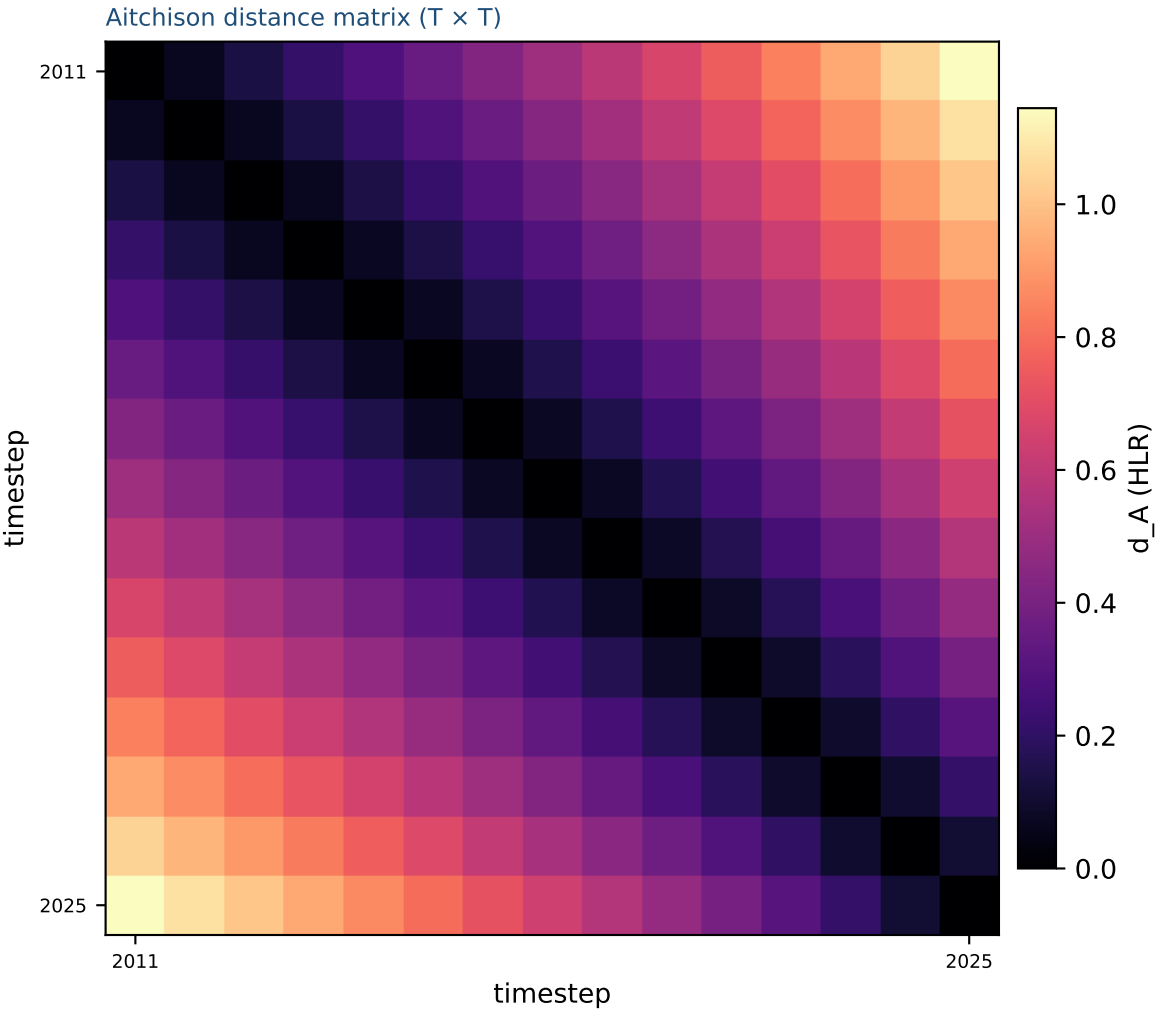
Plate 26 | Pairwise divergence — distance matrix + top-15 ranking

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



rank	A	B	d_A
1	2011	2025	1.1447
2	2012	2025	1.0761
3	2011	2024	1.0374
4	2013	2025	1.0069
5	2012	2024	0.9686
6	2011	2023	0.9377
7	2014	2025	0.9369
8	2013	2024	0.8991
9	2012	2023	0.8686
10	2015	2025	0.8658
11	2011	2022	0.8439
12	2014	2024	0.8288
13	2013	2023	0.7989
14	2016	2025	0.7932
15	2012	2022	0.7746

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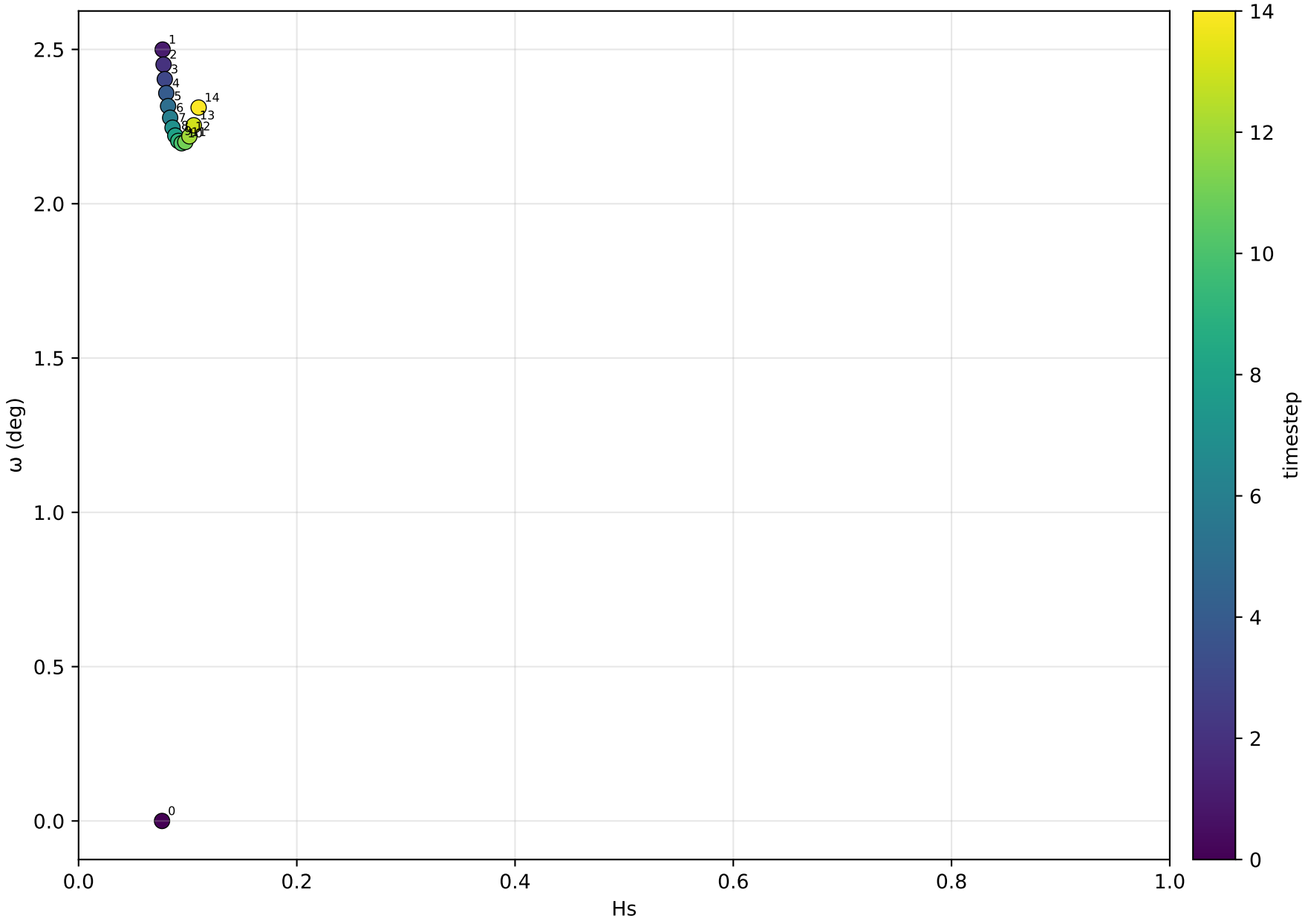
Plate 27 | ω vs Hs scatter (XY test)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



```
T = 15,  D = 8
carriers: Operations & Asset Mgmt, Public Safety, Planning & Building, Recreation & Culture, Library & Heritage, Engineering & Capita

Navigation (System Course Plot):
  net_distance     = 1.1447 HLR
  path_length      = 1.1544 HLR
  course_directness= 0.9916

ω (deg)        : min 0.00  mean 2.14  max 2.50
Hs             : min 0.0765  mean 0.0888  max 0.1100
Aitchison norm : min 1.5391  mean 1.7697  max 2.1155
κ trace        : min 1019.55  max 1916.21
condition #    : min 6.53e+00  max 8.88e+00

Bearings: range [-178.77, 178.49]°,  std 74.60°
Lock events    : 0
Pairs          : 28
|r| > 0.7 pairs: 21

Helmsman frequency (top 5):
  Corporate Services  14  (93.3%)
  (none)           1  (6.7%)

Variation matrix trace : 0.0000
```